PHYSICS CLASS 12 BOARDS 2026

100% Guaranteed High Probability Question Set

☠️ **WARNING: 95% PROBABILITY OF APPEARING IN EXAM**

Based on the pyqs source provided, the questions marked with the "Danger" icon are the highest-probability questions identified in the material. These have been meticulously compiled along with highest-weightage topics for complete coverage.

🔥 95% PROBABILITY QUESTIONS ("DANGER" ICON)

Chapter 1: Electric Charges and Fields

Question: Calculate the electric field and intensity inside a solid sphere of radius R carrying a positive charge with a non-uniform volume charge density $\rho = \rho_0(1 - r/R)$.

> *Context: This involves using Gauss's Law for a solid sphere with variable density.*

Chapter 3: Current Electricity

Question: Derive the expression for the equivalent EMF (ε_{eq}) and internal resistance (r_{eq}) of two batteries connected in **parallel**.

> *Context: Specifically focused on the formula* $\varepsilon_{eq} = (\varepsilon_1 r_2 + \varepsilon_2 r_1)/(r_1 + r_2)$.

Chapter 4: Moving Charges and Magnetism

Question: A 100-turn rectangular coil is hung from a balance scale in a magnetic field. Calculate the additional mass *m* required to regain balance when the magnetic field is switched on.

> *Context: This tests the concept of magnetic force ($F = IlB$) balancing gravitational force.*

Chapter 5: Magnetism and Matter

Question: A magnetic dipole is oriented in three different ways in a uniform magnetic field. Identify which orientation results in the **largest magnetic torque** and which has the **largest potential energy**.

> *Context: Relates to formulas $\tau = mB \sin\theta$ and $U = -mB \cos\theta$.*

Chapter 6: Electromagnetic Induction

Question: Identify the correct graph showing the variation of **induced EMF (ε)** with time for a single-turn coil rotating in a uniform magnetic field.

> *Context: The EMF varies sinusoidally ($\varepsilon = \varepsilon_0 \sin \omega t$). You must pick the correct sine/cosine graph relative to the flux graph.*

Chapter 9: Ray Optics

Question: Derive the Lens Maker's Formula.

> *Context: The question asks you to draw the ray diagram for image formation by a convex spherical surface and then extend it to derive the formula $1/f = (n-1)(1/R_1 - 1/R_2)$.*

Chapter 10: Wave Optics

Question: Analyze the intensity distribution diagram for a **single slit diffraction** pattern. Determine the ratio of the width of the central maximum to the secondary maximum.

> *Context: Conceptual understanding of diffraction fringe widths ($2\lambda D/a$ vs $\lambda D/a$).*

Chapter 13: Nuclei

Question: Study the **Binding Energy per nucleon** curve. Explain the release of energy in nuclear fission and fusion based on the stability of nuclei (mass numbers < 30 and > 170).

> *Context: This is a case-based/passage question connecting Einstein's mass-energy relation to the stability curve.*

🚀 HIGHLY EXPECTED QUESTIONS (FOR REMAINING CHAPTERS)

Chapter 2: Potential and Capacitance

- **Most Important:** Explain the difference in behavior of a **conductor** and a **dielectric** in the presence of an external electric field. Define polarization and susceptibility.
- **Also:** Parallel plate capacitor energy and dielectric breakdown (Case Study).

Chapter 7: Alternating Current

- **Most Important:** A series LCR circuit connected to an ac source. Derive the expression for the **average power** dissipated and define **power factor**. Under what condition is power factor maximum?
- **Also:** Transformer principles and energy losses (Step-up/Step-down).

Chapter 8: Electromagnetic Waves

- **Most Important:** Explain how the **Ampere-Maxwell law** explains the flow of current through a capacitor during charging. Write the expression for **displacement current**.

Chapter 11: Dual Nature of Radiation

- **Most Important:** Experimental study of the **Photoelectric Effect**. Analyze graphs of photoelectric current vs. anode potential for different frequencies and intensities.

Chapter 12: Atoms

- **Most Important: Alpha-particle scattering experiment**. Discuss the trajectory, impact parameter, and distance of closest approach.
- **Also:** Bohr's quantization condition and spectral lines (Lyman, Balmer, etc.).

Chapter 14: Semiconductor Electronics

- **Most Important:** Draw and explain the **V-I characteristics of a p-n junction diode** in forward and reverse bias. Define threshold voltage and breakdown voltage.
- **Also:** Half-wave and Full-wave rectifier diagrams and waveforms.
